Ruiqi Shu 舒睿骐

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Research Interests

My research focuses on machine learning for the physical world, particularly the modeling, simulation, and prediction of dynamical systems. I am interested in differentiable numerical solvers, hybrid physics–learning models, and physically consistent world models. My goal is to connect learning from data with physical laws and numerical computation to model dynamics across spatial and temporal scales. Earth systems, including the atmosphere and oceans, provide concrete settings for this research.

Education

Sep. 2024 – Present

Tsinghua University

PhD Candidate in Atmospheric Science, Department of Earth System Science / Beijing, China

- Advisor: Prof. Xiaomeng Huang.
- Research focus: machine learning for physical dynamical systems, differentiable numerical solvers, and hybrid physics–learning models.

Sep. 2020 – Jun. 2024

Ocean University of China

BSc in Marine Science (Oceanography), Chongben Honors College / Qingdao, China

- Undergraduate training in oceanography, geophysical fluid dynamics, and mathematical modeling.

Research Experience

Nov. 2025 – Present

Shanghai Academy of AI for Science

Research Intern / Shanghai, China

- Conduct research on AI for Science and neural–physics modeling for complex Earth system and ocean dynamics.

Sep. 2024 – Present

Tsinghua University

Graduate Researcher, Department of Earth System Science / Beijing, China

- Develop machine-learning methods that integrate physical constraints, differentiable numerical solvers, and data-driven forecasting.
- Work on hybrid ocean and atmosphere models for long-range simulation, marine heatwave forecasting, and data assimilation.
- Develop Neural-POM, combining a fully differentiable ocean numerical model with machine learning.

Honors & Awards

2026	**National Scholarship**, Ministry of Education of the PRC
2026	**Comprehensive Scholarship**, Tsinghua University
2026	**Selected for the CAST Young Science and Technology Talent Development Program (Doctoral Student Track)**, China Association for Science and Technology
2023	**National 2nd Prize**, Chinese Mathematics Competitions
2022	**National Scholarship**, Ministry of Education of the PRC
2022	**1st Prize**, China Undergraduate Mathematical Contest in Modeling
2019	**Silver Medal**, Chinese Mathematical Olympiad

Publications

1. **HybridOM: Hybrid Physics-Based and Data-Driven Global Ocean Modeling with Efficient Spatial Downscaling**
   Ruiqi Shu, Xiaohui Zhong, Qiusheng Huang, Ruijian Gou, Tianrun Gao, Hao Li, Xiaomeng Huang
   Accepted by ICML 2026 (CCF-A), 2026.
   First author. arXiv

2. **NeuralOM: Neural Ocean Model for Subseasonal-to-Seasonal Simulation**
   Yuan Gao, Hao Wu, Fan Xu, Yanfei Xiang, Ruijian Gou, Ruiqi Shu, Qingsong Wen, Xian Wu, Kun Wang, Xiaomeng Huang
   Proceedings of the AAAI Conference on Artificial Intelligence (CCF-A), pp. 14756–14764, 2026.
   Co-author. DBLP; arXiv; Code

3. **An Exterior-Embedding Neural Operator Framework for Preserving Conservation Laws**
   Huanshuo Dong, Hong Wang, Hao Wu, Zhiwei Zhuang, Xuanze Yang, Ruiqi Shu, Yuan Gao, Xiaomeng Huang
   Accepted by KDD 2026 (CCF-A), 2026.
   Co-author. arXiv

4. **Ocean-E2E: Hybrid Physics-Based and Data-Driven Global Forecasting of Extreme Marine Heatwaves with End-to-End Neural Assimilation**
   Ruiqi Shu, Yuan Gao, Hao Wu, Ruijian Gou, Kun Wang, Yanfei Xiang, Fan Xu, Qingsong Wen, Xiaomeng Huang
   Accepted by KDD 2026 AI4Sciences Track (CCF-A), Oral Presentation (top 6.1%), 2026.
   First author. arXiv

5. **OneForecast: A Universal Framework for Global and Regional Weather Forecasting**
   Yuan Gao, Hao Wu, Ruiqi Shu, Huanshuo Dong, Fan Xu, Rui Ray Chen, Yibo Yan, Qingsong Wen, Xuming Hu, Kun Wang, Jiahao Wu, Qing Li, Hui Xiong, Xiaomeng Huang
   International Conference on Machine Learning (ICML, CCF-A), PMLR 267:18658–18697, 2025.
   Co-first author. PMLR; OpenReview; arXiv; Code

6. **Advanced Forecasts of Global Extreme Marine Heatwaves Through a Physics-Guided Data-Driven Approach**
   Ruiqi Shu, Hao Wu, Yuan Gao, Fanghua Xu, Ruijian Gou, Wei Xiong, Xiaomeng Huang
   Environmental Research Letters, 20(4), 044030, 2025.
   First author. DOI; arXiv

7. **Impact of Downwelling-Favorable Winds on Eddy Formation in the West Greenland Current**
   Ruiqi Shu, Ruijian Gou, Clark Pennelly, Yaocheng Deng, Lixin Wu, Ke Xiao, Yitian Huang, Paul G. Myers
   Journal of Physical Oceanography, 55(2), 191–201, 2025.
   First author. DOI

Publications (continued)

8. **Advancing Ocean State Estimation with Efficient and Scalable AI**
   Yanfei Xiang, Yuan Gao, Hao Wu, Quan Zhang, **Ruiqi Shu**, Xiao Zhou, Xian Wu, Xiaomeng Huang
   arXiv preprint arXiv:2511.06041, 2025.
   Co-author. arXiv

9. **Cracking the Code of Arctic Sea Ice: Why Models Fail to Predict Its Retreat?**
   Ruijian Gou, Gerrit Lohmann, Deliang Chen, Shiming Xu, **Ruiqi Shu**, Shaoqing Zhang, Lixin Wu
   arXiv preprint arXiv:2511.04961, 2025.
   Co-author. arXiv

10. **Advanced Long-Term Earth System Forecasting by Learning the Small-Scale Nature**
    Hao Wu, Yuan Gao, **Ruiqi Shu**, Kun Wang, Ruijian Gou, Chuhan Wu, Xinliang Liu, Juncai He, Shuhao Cao, Junfeng Fang, Xingjian Shi, Feng Tao, Qi Song, Shengxuan Ji, Yanfei Xiang, Yuze Sun, Jiahao Li, Fan Xu, Huanshuo Dong, Haixin Wang, Fan Zhang, Penghao Zhao, Xian Wu, Qingsong Wen, Deliang Chen, Xiaomeng Huang
    arXiv preprint arXiv:2505.19432, 2025.
    Co-first author. arXiv

11. **Turb-L1: Achieving Long-Term Turbulence Tracing by Tackling Spectral Bias**
    Hao Wu, Yuan Gao, **Ruiqi Shu**, Zean Han, Fan Xu, Zhihong Zhu, Qingsong Wen, Xian Wu, Kun Wang, Xiaomeng Huang
    arXiv preprint arXiv:2505.19038, 2025.
    Co-first author. arXiv

12. **BeamVQ: Beam Search with Vector Quantization to Mitigate Data Scarcity in Physical Spatiotemporal Forecasting**
    Weiyan Wang, Xingjian Shi, **Ruiqi Shu**, Yuan Gao, Rui Ray Chen, Kun Wang, Fan Xu, Jinbao Xue, Shuaipeng Li, Yangyu Tao, Di Wang, Hao Wu, Xiaomeng Huang
    arXiv preprint arXiv:2502.18925, 2025.
    Co-author. arXiv

Presentations

Aug. 2026 — **KDD 2026 AI4Sciences Track**
Oral presentation, "Ocean-E2E: Hybrid Physics-Based and Data-Driven Global Forecasting of Extreme Marine Heatwaves with End-to-End Neural Assimilation" / Jeju, South Korea

Jul. 2026 — **International Conference on Machine Learning (ICML 2026)**
Poster presentation, "HybridOM: Hybrid Physics-Based and Data-Driven Global Ocean Modeling with Efficient Spatial Downscaling" / Seoul, South Korea

Jul. 2025 — **International Conference on Machine Learning (ICML 2025)**
Poster presentation, "OneForecast: A Universal Framework for Global and Regional Weather Forecasting" / Vancouver, Canada

Dec. 2024 — **American Geophysical Union 2024 Fall Meeting**
Conference presentation, "A Scale-Aware Framework for Regional to Global Marine Heatwaves Forecast" / Washington, D.C., United States

Academic Service

Reviewer	ICLR, KDD, ICML, AAAI
Research Areas	Physical World Modeling; Dynamical Systems; Differentiable Numerical Solvers; Hybrid Physics–Learning Models